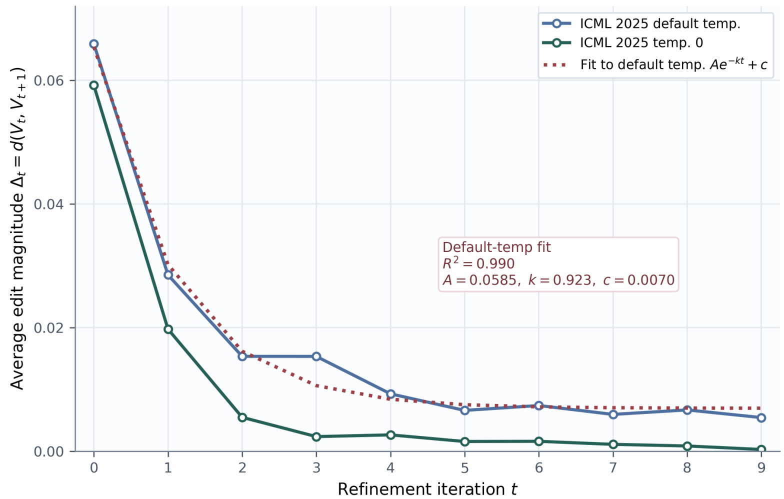

Figure 3: Exponential relaxation of recursive self-refinement. The average transition magnitude $\Delta_t = d(V_t, V_{t+1})$ decays rapidly over early iterations and approaches a small residual floor. The default-temperature trajectory is well fit by $\Delta_t \approx Ae^{-kt} + c$ with $R^2 = 0.990$, suggesting that recursive refinement behaves as a relaxation process toward a soft fixed-point region rather than an open-ended optimization procedure.

Version	Clarity	Conciseness	Technical fidelity	Scientific style	Unnecessary change
V_0	4.0	3.9	5.0	3.5	1.0
V_1	4.7	4.2	5.0	4.1	1.5
V_{10}	5.0	5.0	5.0	5.0	1.6

Table 2: Gemini judge evaluation on 10 sampled ICML 2025 trajectories under temperature 0. Scores are averaged across abstracts on a 1–5 scale. Gemini selects V_{10} as the best version for all 10 abstracts, with 100% meaning preservation.

selects V_{10} as the best version for all 10 abstracts, while assigning perfect technical fidelity and meaning preservation across versions. Average clarity, conciseness, and scientific style all improve from V_0 to V_{10}, although unnecessary-change scores increase slightly, consistent with small residual stylistic churn near convergence.

Cross-year comparison. A smaller evaluation on 15 ICML 2020 abstracts under temperature 0 shows the same qualitative convergence pattern: all trajectories reach both approximate and exact fixed points. ICML 2020 abstracts have slightly larger initial edit distances than ICML 2025 abstracts, 0.0764 versus 0.0592, and slightly later exact convergence, with mean $t^* = 3.00$ versus 2.60. This suggests that the phenomenon is not unique to contemporary abstracts while hinting at possible temporal differences in scientific writing style.

6 Discussion

Our results suggest that recursive LLM refinement is better understood as textual relaxation than as open-ended optimization. Across abstracts and decoding settings, most edits occur in the first few iterations, after which trajectories enter a soft fixed-point region. This behavior is consistent with the view that repeated revision drives text toward a model-preferred region of scientific writing style. Once the text is close to this region, further refinements mostly produce small surface-level changes.

The observed exponential relaxation pattern is particularly noteworthy. Many iterative computational processes exhibit convergence, but the refinement trajectories studied here are well described by a simple exponential decay with a small residual floor. This pattern resembles an overdamped relaxation process: trajectories approach a stable region without evidence of persistent oscillation or divergence.

We do not claim that LLM refinement follows a literal physical law, but the analogy provides a useful language for describing the observed dynamics: rapid early motion, saturation, and residual fluctuations near equilibrium.

This perspective also motivates practical stopping criteria. Although recursive refinement is sometimes described as an optimization-like procedure, our results suggest that its useful progress can be monitored directly through the residual edit magnitude $\Delta_t = d(V_t, V_{t+1})$. A simple stopping rule is to terminate refinement once Δ_t remains below a small threshold for consecutive iterations, optionally combined with word-count stability. This is analogous in spirit to convergence criteria used in iterative optimization, but does not require assuming that the model is explicitly performing gradient descent on a known objective.

The distinction between exact and approximate fixed points is important. Exact textual equality is a useful but overly strict criterion: a model may continue to adjust punctuation, hyphenation, or local phrasing even when the text is essentially stable. Approximate convergence better captures the practical stopping point of recursive refinement. In our experiments, approximate convergence occurs earlier and more consistently than exact convergence, suggesting that users may need only a small number of refinement rounds before additional iterations yield diminishing returns.

Temperature affects the nature of the fixed-point region. Deterministic decoding with temperature 0 quickly collapses to exact fixed points, while default-temperature decoding exhibits residual fluctuations around a similar soft fixed-point region. This supports the interpretation of temperature as controlling local variability near convergence rather than changing the overall direction of refinement.

More broadly, textual relaxation suggests that recursive LLM workflows may admit a richer stability analysis beyond surface edit distance. In a continuous representation space, one could study the local stability spectrum of refinement near a soft fixed-point region, characterizing which textual directions contract rapidly and which remain variable under repeated revision. Such an analysis could connect empirical edit-magnitude saturation to a more formal account of stability in recursive model use. It may also clarify whether refinement reduces trajectory-level variability as texts approach the fixed-point region.

These findings have practical implications for LLM-assisted writing workflows. Repeated refinement is not necessarily harmful, but it appears to have rapidly diminishing returns. For scientific abstracts, a few iterations are usually sufficient to reach a stable and improved version. Beyond that point, additional iterations may mostly introduce stylistic churn. This suggests that refinement systems should monitor convergence metrics and stop automatically once changes fall below a soft fixed-point threshold.

7 Limitations

Our study is limited in scope. We focus primarily on scientific abstracts from ICML proceedings and use GPT-5.5 as the main refinement model. The observed convergence behavior may differ for other genres, longer documents, more creative writing tasks, code refinement, multi-turn agentic workflows, or models with substantially different decoding behavior. Our cross-year comparison with ICML 2020 abstracts is smaller than the main ICML 2025 experiment and should be interpreted as a preliminary validation rather than a comprehensive temporal analysis.

Our convergence metrics are surface-based. Normalized edit distance and word-count stability are useful for measuring textual change, but they do not fully capture semantic equivalence, scientific correctness, or deeper rhetorical structure. We partially address this limitation with an external Gemini judge, but LLM-as-a-judge evaluation is itself imperfect and may reflect model-specific preferences. Human expert evaluation would provide a stronger assessment of whether refined abstracts preserve all technical claims while improving scientific communication.

The relaxation model is empirical. Although the exponential fit provides a compact description of the observed dynamics, it should not be interpreted as a physical law or as evidence that LLM refinement literally performs gradient descent on an explicit objective. The analogy to relaxation is intended to describe the observed shape of the refinement trajectory: rapid early change, saturation, and a small residual floor. Establishing a mechanistic theory would require stronger evidence, such as identifying an explicit objective, estimating local contraction properties, or analyzing the refinement operator in a continuous representation space.

Finally, our dynamical-systems analysis remains preliminary. We do not yet characterize the local stability spectrum of the refinement process in continuous representation spaces, nor do we directly analyze trajectory-level variability beyond surface edit distance. Such analyses are important directions for future work toward a fuller theory of textual relaxation.

8 Conclusion

We studied recursive self-refinement in large language models as *textual relaxation*: a discrete dynamical process over text induced by repeated application of the same revision operator. Using GPT-5.5 refinement trajectories on ICML abstracts, we found that repeated self-refinement rapidly saturates. Most edits occur in the first few iterations, after which trajectories enter a soft fixed-point region with only small residual changes. Deterministic decoding reaches exact fixed points faster and with lower residual fluctuation than default-temperature decoding, while both settings achieve universal approximate convergence. The average edit magnitude is well described by an exponential relaxation curve, suggesting that recursive refinement behaves more like relaxation toward a model-preferred textual equilibrium than open-ended optimization.

These results suggest that approximate convergence metrics are useful for understanding and controlling iterative LLM workflows. In practical writing systems, monitoring edit distance and length stability can provide a simple stopping rule, preventing unnecessary refinement once the text has reached a stable region. More broadly, viewing LLM self-refinement through the lens of dynamical systems offers a compact framework for studying convergence, stability, residual variation, and soft fixed-point behavior in recursive model use. We hope this perspective motivates further work on the dynamics of iterative AI systems, including broader task domains, multi-model comparisons, and representation-space analyses of textual relaxation.

References

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522, 2023.

Aman Madaan, Niket Tandon, Peter Clark, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik R. Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023.